

????? ?? Dependency Injection ? Zenject

Unity-??????? : SlimeRush. ?????: Dependency Injection, Zenject, SceneContext ? ProjectContext.

1. ????? ???????

???? ??????? - ??????? Dependency Injection ? Unity ? ????????????????? Zenject, ? ?????? ?????????????, ??? ????????????????? ????????????? ??????? installers, SceneContext ? ProjectContext.

? ??????? SlimeRush ????? ??????? ????????????????? ????????????? ??????? ????????. ?????? ????????????????? ?????? UI, ????????? ?????? ? ?????????? ?????????? ????? ??????????? ?? ??????????????? ?? ???????????????.

?????: ?? ????????? ?????? ? ?????????? ??? ????????????????? Zenject ?????? SceneContext. ????????????????? ProjectContext ? ????????????????? ?????????????? ?????????????? ?????????????? ?????????????? ?????????????? ?????????????? ?????????????? ?????????????? ??????????????.

2. ????????????? Zenject

? ??????? ??? ?????????? Zenject. ?? ?????? GameplayPrototype ????????????????? SceneContext, ? ?????????? ????????????? GameplayInstaller. ?????? installer ?????????????????? ??????????????, ??????????????? ??? ?????????? ???????.

```
Container.BindInstance(crowdFormationSettings).AsSingle();
Container.Bind<ICrowdFormation>().To<EllipseCrowdFormation>().AsSingle();

Container.Bind<SlimeCrowdManager>().FromInstance(slimeCrowdManager).AsSingle();
Container.Bind<ISlimeCrowd>().FromInstance(slimeCrowdManager).AsSingle();
Container.Bind<ISlimeCrowdCommands>().FromInstance(slimeCrowdManager).AsSingle();
```

??? ?????????????????? ?????????????????? ?????????????????????? ?????? ?????????????? Zenject, ? ?? ?????????? ?????????? ?????????? ??????????.

3. ?????????????????? ??????????????

??? ?????????? ?????????? ?????? ?????????? ?????????????? ??????????????: ????? ??? ?????????? ?????????????? ???????, ?????????? ??? ?????????, ?????????????????? ?????????????.

```
public interface ISlimeCrowd
{
    int SlimeCount { get; }
    event Action<int> OnSlimeCountChanged;
}

public interface ISlimeCrowdCommands
{
    void AddSlimes(int amount);
    void RemoveSlimes(int amount);
    void MultiplySlimes(int multiplier);
}
```

?????? ?????????????????? ?????????????????? Interface Segregation Principle. UI ?????????? ??????? ISlimeCrowd ? ?? ?????? ?????????????? ?????????????? ??????????. ?????????? ??????????, ?????????? ?????????, ?????????? ??? ??????, ?????????? ?????????????????? ISlimeCrowdCommands.

4. ?????????????? ?????????? ??????????

?? ?????????????? SlimeCrowdManager ??????? ?? ??????? ?? ????????? ? ????????? ?????????, ??
? ?? ?????????????????? ??????? ?? ??????????????. ??? ?????????? ?????????? ??????????????
????????????????????.

```
public interface ICrowdFormation
{
    IReadOnlyList<Vector3> GeneratePositions(int count);
}

public class EllipseCrowdFormation : ICrowdFormation
{
    private readonly CrowdFormationSettings settings;

    public EllipseCrowdFormation(CrowdFormationSettings settings)
    {
        this.settings = settings;
    }
}
```

??????? SlimeCrowdManager ?? ?????, ??? ?????? ?????????????????? ?????? ??????. ?? ??????????
ICrowdFormation ?????? Zenject ? ?????????????? ??????? ?????? GeneratePositions.

```
[Inject]
private void Construct(ICrowdFormation crowdFormation)
{
    this.crowdFormation = crowdFormation;
}
```

5. ScriptableObject ??? ?????????

??? ?????????????? ?????????? ??? ??????? CrowdFormationSettings : ScriptableObject. ? ????
????????????? ??????????, ?????????? ?? ?????? ??????:

- ? baseSpacing
- ? ellipseWidth
- ? ellipseDepth
- ? frontOffsetZ
- ? randomOffsetAmount

ScriptableObject ??????? ??? Unity, ??????? ??? ?????????????? ?????? ??????? ?????? Inspector
??? ?????????????? ??????. ?????? ?????? ?????????????? ??????? assets ? ?????????????????? ??? ???????
??????????.

6. Dependency Injection ? UI

????????????? CrowdCountLabel ?????????? ?????????????????? ?????? Zenject:

```
[Inject]
private void Construct(ISlimeCrowd slimeCrowd)
{
    this.slimeCrowd = slimeCrowd;
}
```

????????????? ?????? UI ?? ?????????? ?????????? ?? SlimeCrowdManager. ?? ?????????? ??????? ?
????????????????? ISlimeCrowd, ?? ?????? ??????? ??????? ?????????????.

7. ProjectContext ? ?????????????????? ?????????

?? ?????????? ?????????????????? ?????????? ProjectContext ? ?????????? ? ?????? ?????????????????? ??????????
?????? ?? ?????????????????? ?????? ??? ????????. ProjectContext ?????????? ?????????????????? ???????
?????????????????, ?????????? ??????? ?? ?????????????????? ?????? ?????? ? ?? ?????????????????? ? ??????????????????

??????.

??? SlimeRush ?????????? ?????????? ????? ????:

? IInputService - ?????? ?????????? ?????.

? ISaveSystem - ?????????? ? ?????????? ??????????.

? ISceneService - ?????????????????? ?????????? ????

? IAudioService - ?????????? ?????????? ?????.

? IGameProgressService - ?????????? ?????? ????? ?????????.

?? ?????????? ?????????? ? ProjectContext ?????????? ?????????????? ??????: SlimeCrowdManager,

??????, ??????, level spawner ??? ?????????? ?????????? ?????????????? ??????. ??? ??????????????

????????? ?????????????? ? SceneContext.

8. ?????????????? ProjectInstaller

```
public class ProjectInstaller : MonoInstaller
{
    public override void InstallBindings()
    {
        Container.Bind<IInputService>().To<InputService>().AsSingle();
        Container.Bind<ISaveSystem>().To<SaveSystem>().AsSingle();
        Container.Bind<ISceneService>().To<SceneService>().AsSingle();
        Container.Bind<IAudioService>().To<AudioService>().AsSingle();
    }
}
```

????? installer ?????? ??? ? ?????????? ? ?????????? ProjectContext. ?????? ?????? ?????????

?????? ?????????? ?? ??? ? ?????? ?????????.

9. ?????? Lifetime

??? ?????????????? ?????????? ?????? ?????????????????? AsSingle, ??????? ??? ??? ???????

????????????????? ? ?????? ?????????????? ?? ?????????????? ????? ?????.

? InputService - AsSingle: ??? ? ?????????? ?????? ??? ????? ?????.

? SaveSystem - AsSingle: ?????? ?????????? ?????????????.

? SceneService - AsSingle: ?????? ?????? ?????????? ????

? AudioService - AsSingle: ??? ? ?????????????? ?????????????? ??????

AsTransient ?? ?????????? ??? ?????? ??????????, ??????? ??? ?????? ?????????? ?????????? ?? ??????

?????????. ??? ?????? ?????????? ? ??????? ?????????????? AsCached ?????? ?????????????????? ???

?????????? ?????????? ?????????? ??? ??????? ?????????, ?? ??? ?????????? ?????????????? ??????????

AsSingle ?????? ? ??????????

10. ?????????????????? Dependency Injection

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? UI ?? ?????????? ??????? ?????? ? ?????????? ???????.

? ?????????????????? ?????????????? ?????????? ?????? ?????? MonoBehaviour.

? ?????????????????? ??????? ?????? ??? ? ?????? Construct ??? constructor.

? ??? ?????????????? ?????? ????????????????? ? ?????????????.

? ?????????????????? ?????????????? ?????? ? ?????????????? ?????????? ?????.

11. ??? ?????? ?????????????? ? ??????

? Unity-??????? ? ?????????????????????? Zenject.

? ??????????? ? ?????????????????? ProjectContext.

? ProjectInstaller ? ?????????????????? ?????????????.

? ??????? ?????????????????????? ??? ? ? ?????? ?????????? ?????? [Inject].

? ?????????? ?????????? ProjectContext ? ProjectInstaller.
? PDF-????? ? ?????????? ?????????????????? ?????????.

12. ??????

? ???????? SlimeRush ??? ?????????????? Zenject ??? ?????????????? ?????????????????? ?? ???????
?????. ??? ? ?????????? ?????????????? ??? ?????? ? ?????????? ?????????? ??????, ? ??????
????????? ??? ???????? ? ?????????? ?????????????? ICrowdFormation.
??????????? ??? ??? ?????????? ?????????????? ???????? - ?????????? ProjectContext ?
????????????????????? ? ??? ?????????????? ?????????? ?????????? ??? ??? ? ?????????? ??????????
????????????????? ??????? ??????? ?????????????? ? SceneContext, ??? ??? ??? ?????????????? ?
????????????????? ?????????? ???????.